

Challenging Minesweeper By Noam Kyrar & Ron Eliav

This game is based on the known game Minesweeper with a challenging twist. The purpose of the game is to detect all the mines on the board. Every cell on the board that is not a bomb has a number of adjacent bombs. But when the player finds a free cell bomb it only reveals the adjacent cells with zero adjacent bombs. To play the game the user moves with the arrows, marks a cell with a flag with 'M', and selects a bomb-free cell with space.

Architecture

The game contains 4 Jack classes:

Main.jack -

This Class manages the current game out of 3. In every game, the method creates a Game object which is responsible for creating the handling the current game. After every iteration, we clear the screen and dispose of the game object.

GameScreen.jack -

This class has several functions each of them draws another part - printing the board's frame, the title, all the numbers, the square cursor, the flag and bomb icon, the instruction, and the game-over and victory message.

Game.jack:

The game class has 3 fields: Board in the type of Array of Arrays which keeps the original board, Discover board which keeps the current board which is shown on the screen, numBoard which keeps the current game.

Constructor -

It initializes the 2 boards fields.

StartGame -

The main loop of the game. first calls to printScreen and then enter the loop:

1. check if any key is pressed, if so the method keyPressed is called checks which key was pressed.
2. If the space key is pressed it calls the method step. which checks if the current cell has a bomb. If so, the game is over. Otherwise, the function copySquareArray is called - and all the adjacent cells of the current cell are copied to the discoverBoard. If the flag key is pressed mark the current cell with a flag in the discover board. To remove the flag just press again.
3. The loop is done in case of a victory or falling to a bomb.

checkWin -

It iterates the both boards and checks if there are no remaining number cells in the original board which is not in the discover board.

dispose -

This method disposes the memory block of the current Game object.

fillBorads -

Creates 3 Arbitrary boards manually.

Square.Jack

This class has 6 fields: X and Y coordinates which keeps the current location on the screen, sizeX and sizeY keep the width and length of every cell, row and col which keeps the indices in the board.

Constructor-

Initializes the fields and puts the first location of the cursor in the middle square of the board.

getRow And getCol

dispose -

This method disposes the memory block of the current Game Square.

Draw -

This function draws the cursor on the screen.

Erase -

Erase the cursor to make it blink.

Move functions -

Changes the fields according to the pressed arrow.

The idea of creating this challenging edition of Minesweeper came after having hours of fun in the original game until it became too easy. Then we needed to think about a more challenging version and keep the game as interactive and fun as the original game.

Video Demo Clip:

[Minesweeper Demo](#)

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